ETCS Level 2 Corridor Upgrade (142 km double track)

Project owner	Signalling & Control-Command	**Budget (CAPEX)**	€ 38,000,000
Planned start	Q2 2027	**Planned completion**	Q2 2029

1. Project description

Migration of a 142 km mixed-traffic double-track corridor from national class-B signalling to ETCS Level 2 with GSM-R/FRMCS-ready radio. Includes two new electronic interlockings, one Radio Block Centre, trackside train detection renewal and on-board ETCS retrofit of 24 locomotives. Works are executed in night possessions; commissioning in three stages with fallback to legacy signalling until final cut-over.

2. Objectives

- Increase line capacity by 20%
- Comply with EU ERTMS Deployment Plan (TEN-T core network)
- Remove 310 lineside signals
- Retrofit 24 locomotives with ETCS Baseline 3

3. Bill of materials

#	Item ID	Description	Standard	Qty	Supplier ID	Supplier
01	007-001	Radio Block Centre (RBC), redundant 2oo3	EN 50126/8/9 SIL4	1 pc	SUPP_01536	Advanced Manufacturing 760
02	007-002	Electronic interlocking, object controller cabinet	EN 50129 SIL4	2 pc	SUPP_03583	Apex Manufacturing 188
03	007-003	Eurobalise, fixed, with mounting kit	SUBSET-036	1,240 pc	SUPP_04968	Prime Industries 497
04	007-004	Lineside Electronic Unit (LEU)	SUBSET-047	60 pc	SUPP_08281	Global Systems 623
05	007-005	Axle counter detection point	EN 50617-2	420 pc	SUPP_09865	Global Systems 220
06	007-006	Point machine, electro-hydraulic	EN 50126	96 pc	SUPP_00435	Strategic Industries 893
07	007-007	GSM-R base transceiver station	EIRENE SRS	38 pc	SUPP_09197	Prime Manufacturing 492
08	007-008	Fibre-optic cable 48F, armoured	IEC 60794	300 km	SUPP_03258	Prime Group 644
09	007-009	On-board ETCS kit (EVC, DMI, BTM antenna, odometry)	SUBSET-026 B3	24 set	SUPP_08930	Global Technologies 968
10	007-010	Signalling power supply, UPS 10 kVA	EN 50155	64 pc	SUPP_06875	Dynamic Technologies 461
11	007-011	Lineside equipment cabinet, outdoor IP55	EN 50125-3	140 pc	SUPP_03612	Advanced Technologies 117
12	007-012	Data cable 4×1.5 mm² signalling, LSZH	EN 50288	420 km	SUPP_07361	Elite Logistics 262

4. Procurement notes

- Suppliers listed are the current preferred source per item; alternatives may be proposed after risk review.
- All items require ISO/TS 22163 (IRIS) or equivalent certification and first article inspection before series delivery.
- Supplier due diligence per LkSG / CSDDD is required before purchase order release.

SIMULATED DEMO DOCUMENT – fictitious project for hackathon use. Supplier IDs and names refer to supplier_risk_dataset.csv.